Super-twisting lecture notes

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March 2026

1 Introduction

1.1 By the end of this lecture

- You should have an overview of super-twisting sliding mode control, strengths, weaknesses, variations and solutions

1.2 Sliding mode recap

Second-order system

$$\begin{aligned}\dot{x}_1 &= x_2 \\ \dot{x}_2 &= h(x) + g(x)u\end{aligned}$$

where h and g are unknown nonlinear functions of time and $g(x) \geq g_0 > 0, \forall x$. If we want to stabilize the origin we can choose

$$s = \lambda x_1 + x_2$$

with derivative

$$\dot{s} = \lambda x_2 + h(x) + g(x)u$$

if $s = 0$ we obtain our sliding manifold (surface). This gives the dynamics

$$x_2 = -\lambda x_1$$

Meaning that the dynamics go exponentially to zero on the sliding manifold.

Therefore, we want to choose a control law that sends the trajectory to the sliding manifold and stays there, i.e. $s = 0$ and $\dot{s} = 0$. Through Lyapunov analysis we end up with the function

$$u = -\beta(x) sgn(s)$$

where

$$sgn(s) = \begin{cases} 1, & s > 0 \\ 0, & s = 0 \\ -1, & s < 0 \end{cases}$$

2 Adaptive super-twisting algorithm

2.1 Intro

Key problems to solve

1. Chattering
2. Control gain uncertainty and overestimation
3. Input gain uncertainty

2.1.1 System

Given a single-input uncertain nonlinear system on the form

$$\dot{x} = f(x,t) + h(x,t)u \tag{1}$$

where $x \in \mathbb{R}^n$ is the state vector, $u \in \mathbb{R}$ is a control function, $f(x,t) \in \mathbb{R}$ is a differentiable, partially know vector-field.

2.1.2 Assumptions

Assumption 1. *A sliding variable $\sigma = \sigma(x,t) \in \mathbb{R}$ is designed so that the system's (1) desirable compensated dynamics are achieved in the sliding mode $\sigma = \sigma(x,t) = 0$.*

Assumption 2. *The system's (1) input-output ($u \to \sigma$) dynamics are of a relative degree one, and the internal dynamics are stable. Therefore, the input-output dynamics can be presented*

$$\begin{aligned} \dot{\sigma} &= \underbrace{\frac{\partial \sigma}{\partial t} + \frac{\partial \sigma}{\partial x} f(x,t)}_{\phi(x,t)} + \underbrace{\frac{\partial \sigma}{\partial x} h(x)u}_{b(x,t)} \\ \dot{\sigma} &= \phi(x,t) + \omega, \quad \omega = b(x,t)u \iff u = b^{-1}(x,t)\omega \end{aligned} \tag{2}$$

*The solution of system (2) is understood in the sense of Filippov [**?**].*

Remark 1. *About Filippov*

Assumption 3. *Further, assume that function $b(x,t) \in \mathbb{R}$ is known and $b(x,t) \neq 0$ $\forall x$ and $t \in [0,\infty)$.*

Assumption 4. *the perturbation function $\phi(x,t) \in \mathbb{R}$ is bounded*

$$|\phi(x,t)| \leq \delta |\sigma|^{1/2} \tag{3}$$

where the finite boundary $\delta > 0$ exists but is not known.

2.2 The adaptive STW controller

2.2.1 Controller

The STW controller as written by Shtessel et al. [**?**] is given by

$$\begin{aligned} \omega &= -\alpha |\sigma|^{1/2} sgn(\sigma) + v \\ \dot{v} &= -\beta sgn(\sigma) \end{aligned} \tag{4}$$

where $\alpha = \alpha(\sigma, \dot{\sigma}, t)$ and $\beta = \beta(\sigma, \dot{\sigma}, t)$ are adaptive gains. The closed-loop sliding dynamics are then on the form

$$\begin{aligned} \dot{\sigma} &= -\alpha |\sigma|^{1/2} sgn(\sigma) + v + \phi(x,t) \\ \dot{v} &= -\beta sgn(\sigma) \end{aligned} \tag{5}$$

2.2.2 Adaptive gains

Theorem 1 (Shtessel et al. [**?**])**.** *Consider system (5). Suppose that the perturbation $\phi(x,t)$ satisfies (3) for some unknown constant $\delta > 0$. Then for any initial conditions $x(0), \sigma(0)$ the sliding surface $\sigma = 0$ will be reached in finite time via STW control (4) with the adaptive gains*

$$\begin{aligned} \dot{\alpha} &= \begin{cases} \omega_1 \sqrt{\frac{\gamma_1}{2}}, & \text{if } \sigma \neq 0, \\ 0, & \text{if } \sigma = 0 \end{cases} \\ \beta &= 2\epsilon\alpha + \lambda + 4\epsilon^2 \end{aligned} \tag{6}$$

where $\epsilon, \lambda, \gamma_1$ and ω_1 are arbitrary positive constants.

2.2.3 Proof of Theorem 1

Proof in 2 steps

1. Change of variable
2. Lyapunov stability analysis to prove finite time convergence to the sliding manifold

Proof. To simplify the Lyapunov analysis we perform a change of variable

$$z = \begin{bmatrix} z_1 \\ z_2 \end{bmatrix} = \begin{bmatrix} |\sigma|^{1/2} sgn(\sigma) \\ v \end{bmatrix}$$

then rewrite (5) to

$$\begin{cases} \dot{z}_1 &= \frac{1}{|z_1|}\left(-\frac{\alpha}{2} z_1 + \frac{1}{2} z_2 + \frac{1}{2}\phi(x,t)\right) \\ \dot{z}_2 &= -\frac{\beta}{|z_1|} z_1 \end{cases} \tag{7}$$

Equation (7) can then be written on matrix form

$$\dot{z} = \underbrace{\frac{1}{|z_1|}\begin{bmatrix} -\frac{\alpha}{2} & \frac{1}{2} \\ -\beta & 0 \end{bmatrix}}_{A(z_1)} \begin{bmatrix} z_1 \\ z_2 \end{bmatrix} + \underbrace{\frac{1}{|z_1|}\begin{bmatrix} \frac{1}{2} \\ 0 \end{bmatrix}}_{g(z_1)} \phi(x,t) \tag{8}$$

Remark 2 (Observe)**.** *We observe the following*

1. *If $z_1, z_2 \to 0$ in finite time then $\sigma_1, \sigma_2 \to 0$ in finite time, and*
2. *$|z_1| = |\sigma|^{1/2}$ and $sgn(z_1) = sgn(\sigma)$*

Lyapunov analysis: In the second step of the proof, the stability analysis of the system given by the above equation is performed. We introduce the LFC

$$V(z_1, z_2, \alpha, \beta) = V_0 + \frac{1}{2\gamma_1}(\alpha - \alpha^*)^2 + \frac{1}{2\gamma_2}(\beta - \beta^*)^2 \tag{9}$$

where

$$V_0(z) = \left(\lambda + 4\epsilon^2\right) z_1^2 + z_2^2 - 4\epsilon z_1 z_2 = z^\top P z \tag{10}$$

$$P = \begin{bmatrix} \lambda + 4\epsilon^2 & -2\epsilon \\ -2\epsilon & 1 \end{bmatrix}, \ \forall \lambda, \epsilon > 0.$$

where $\alpha^*, \beta^* > 0$ are some constants.

The derivative of the Lyapunov function candidate is

$$\dot{V}(z, \alpha, \beta) = \dot{z}^\top P z + z^\top P \dot{z} + \frac{1}{\gamma_1}(\alpha - \alpha^*)\dot{\alpha} + \frac{1}{\gamma_2}(\beta - \beta^*)\dot{\beta} \tag{11}$$

We can bound the first two terms

$$\dot{V}_0 = \dot{z}^\top P z + z^\top P \dot{z} \leq -\frac{1}{|z_1|} z^\top \tilde{Q} z \tag{12}$$

where the symmetric matrix $\tilde{Q}$ is given by

$$\tilde{Q} - 2\epsilon I = \begin{bmatrix} 2\lambda\alpha + 4\epsilon(2\epsilon\alpha - \beta) - 2\left(\lambda + 4\epsilon^2\right)\delta & * \\ \left(\beta - 2\epsilon\alpha - \lambda - 4\epsilon^2\right) + 2\epsilon\delta & 4\epsilon \end{bmatrix} \tag{13}$$

To ensure positive definite $\tilde{Q}$ we enforce

$$\beta = 2\epsilon\alpha + \lambda + 4\epsilon^2 \tag{14}$$

The matrix $\tilde{Q}$ is positive definite with a minimal eigenvalue $\lambda_{\min}(\tilde{Q}) \geq 2\epsilon$ if

$$\alpha > \frac{\epsilon\delta^2 + \left(\lambda + 4\epsilon^2\right)(2\epsilon + \delta) + \epsilon}{\lambda} \tag{15}$$

We then have that

$$\dot{V}_0 \leq -\frac{1}{|z_1|} z^\top \tilde{Q} z \leq -\frac{\epsilon}{|z_1|} z^\top z = -\frac{\epsilon}{|z_1|} |z|^2 \tag{16}$$

We also have that

$$\lambda_{\min}(P) |z|^2 \leq z^\top P z \leq \lambda_{\max}(P) |z|^2 \tag{17}$$

where $|z|^2 = z_1^2 + z_2^2 = |\sigma| + z_2^2$ and

$$|z_1| = |\sigma|^{1/2} \leq |z|^2 \leq \frac{V_0^{1/2}(z)}{\lambda_{\min}^{1/2}(P)} \tag{18}$$

then

$$\dot{V}_0(z) \leq -r V_0^{1/2}, \; r = \frac{2\epsilon \lambda_{\min}^{1/2}(P)}{\lambda_{\max}(P)} \tag{19}$$

We also have that

$$\begin{aligned}
\dot{V}(z,\alpha,\beta) &= \dot{z}^\top P z + z^\top P \dot{z} + \frac{1}{\gamma_1}(\alpha - \alpha^*)\dot{\alpha} + \frac{1}{\gamma_2}(\beta - \beta^*)\dot{\beta} \\
&\leq -\frac{1}{|z_1|} z^\top \tilde{Q} z + \frac{1}{\gamma_1}(\alpha - \alpha^*)\dot{\alpha} + \frac{1}{\gamma_2}(\beta - \beta^*)\dot{\beta} \\
&\leq -r V_0^{1/2} + \frac{1}{\gamma_1}(\alpha - \alpha^*)\dot{\alpha} + \frac{1}{\gamma_2}(\beta - \beta^*)\dot{\beta} \\
&= -r V_0^{1/2} - \frac{\omega_1}{\sqrt{2\gamma_1}} |\alpha - \alpha^*| \dot{\alpha} - \frac{\omega_2}{\sqrt{2\gamma_2}} |\beta - \beta^*| \\
&+ \frac{1}{\gamma_1}(\alpha - \alpha^*)\dot{\alpha} + \frac{1}{\gamma_2}(\beta - \beta^*)\dot{\beta} + \frac{\omega_1}{\sqrt{2\gamma_1}} |\alpha - \alpha^*| + \frac{\omega_2}{\sqrt{2\gamma_2}} |\beta - \beta^*|
\end{aligned} \tag{20}$$

Using the inequality

$$\left(x^2 + y^2 + z^2\right)^{1/2} \leq |x| + |y| + |z| \tag{21}$$

using this for the Lyapunov candidate function

$$-r V_0^{1/2} - \frac{\omega_1}{\sqrt{2\gamma_1}} |\alpha - \alpha^*| \dot{\alpha} - \frac{\omega_2}{\sqrt{2\gamma_2}} |\beta - \beta^*| \leq -\eta V(z,\alpha,\beta)^{1/2} \tag{22}$$

where $\eta = \min(r, \omega_1, \omega_2)$.

Using equation (22) we can rewrite the upper bound on $\dot{V}$

$$\begin{aligned}
\dot{V}(z,\alpha,\beta) \leq -\eta V(z,\alpha,\beta)^{1/2} &+ \frac{1}{\gamma_1}(\alpha - \alpha^*)\dot{\alpha} + \frac{1}{\gamma_2}(\beta - \beta^*)\dot{\beta} \\
&+ \frac{\omega_1}{\sqrt{2\gamma_1}} |\alpha - \alpha^*| + \frac{\omega_2}{\sqrt{2\gamma_2}} |\beta - \beta^*|
\end{aligned} \tag{23}$$

Assuming the adaptation law (6) makes the adaptive gains $\alpha(t)$ and $\beta(t)$ bounded there exists positive constants α^* and β^* such that $\alpha(t) - \alpha^* > 0$ and $\beta(t) - \beta^* > 0$, $\forall t \geq 0$. Using this assumption we can reduce the upper bound expression to

$$\dot{V}(z,\alpha,\beta) \leq -\eta V(z,\alpha,\beta)^{1/2} + \xi \tag{24}$$

where

$$\xi = -|\alpha - \alpha^*| \left(\frac{1}{\gamma_1}\dot{\alpha} - \frac{\omega_1}{\sqrt{2\gamma_1}} \right) - |\beta - \beta^*| \left(\frac{1}{\gamma_2}\dot{\beta} - \frac{\omega_2}{\sqrt{2\gamma_2}} \right) \tag{25}$$

We want to ensure $\xi = 0$

$$\dot{\alpha} = \omega_1 \sqrt{\frac{\gamma_1}{2}} \tag{26}$$

$$\dot{\beta} = \omega_2 \sqrt{\frac{\gamma_2}{2}} \tag{27}$$

If we choose $\epsilon = \frac{\omega_2}{2\omega_1}\sqrt{\frac{\gamma_2}{\gamma_1}}$ and differentiating β as given by (6) we get

$$\dot{\beta} = 2\epsilon\dot{\alpha} = 2\epsilon\omega_1\sqrt{\frac{\gamma_1}{2}} = \omega_2\sqrt{\frac{\gamma_2}{2}} \tag{28}$$

Observe. For the controller to converge in finite time the inequality (15). This implies that $\alpha(t)$ must increase according to (26) until (15) is met in order to guarantee a positive definite $\tilde{Q}$ matrix.

> Also, as soon as the sliding variable σ and its derivative converges to zero it does not make sense to increase $\alpha(t)$ and $\beta(t)$ by making $\dot{\alpha} = 0$ as $\sigma = 0$. - Shtessel et al. [**?**]

Hence, we obtain the adaptation law (6). □ □

2.3 Propositions

Proposition 1. *The adaptive gains $\alpha(t)$ and $\beta(t)$ are bounded.*

Proof. A solution to the adaptive gain $\alpha(t)$ in (6)

$$\alpha = \begin{cases} \alpha(0) + \omega_1\sqrt{\frac{\gamma_1}{2}}t, & 0 \le t \le t_r \\ \alpha(0) + \omega_1\sqrt{\frac{\gamma_1}{2}}t_r, & t > t_r \end{cases} \tag{29}$$

Hence, $\alpha(t)$ and $\beta(t)$ are bounded. □ □

Proposition 2. *The finite time t_r can be estimated as*

$$t_r \le \frac{2V^{1/2}(0)}{\eta} \tag{30}$$

where $\eta = min(r, \omega_1, \omega_2)$.

Proof. Since the right-hand side of the inequality (15) is bounded and $\alpha(t)$ is increasing linearly with respect to time according to the adaptation law (6), it is met in finite time. The above inequality is found by integrating (24). □ □

Key takeaways

- Deals well with matched uncertainties
- Chattering is attenuated by having the switching term behind an integrator
- Adaptive gain law is obtained and drives the system to the desired state in finite time
- The input gain is assumed known
- Controller gains are increasing
- Many design variables with tuning lack of intuition

3 Generalized super-twisting algorithm

3.1 Motivational example

3.1.1 Velocity-tracking problem

$$\left(1 + \cos^2(q)\right)\ddot{q} + g\sin(q) + b\left(\dot{q} + \arctan(\dot{q})\right) = u$$

where $q \in \mathbb{R}$ and $\dot{q} \in \mathbb{R}$ are state variables and $u \in \mathbb{R}$ is the torque control input

3.1.2 Solution

$$\begin{aligned}\dot{e}_2 = &- k_2\lfloor e_1\rceil^0 + \dot{\varphi}\underbrace{-bc\gamma k_1\lfloor e_1\rceil^{1/2} - bc\gamma k_2\int_0^t \lfloor e_1(\tau)\rceil^0 d\tau}_{u(t)} \\ &+ bc\gamma\int_0^t \left[\dot{\varphi}(\tau) - b\left(k_1\lfloor e_1(\tau)\rceil^{1/2} - e_2(\tau)\right)\right] d\tau\end{aligned} \tag{31}$$

3.2 Generalized super-twisting algorithm

3.2.1 Control law

The generalized super-twisting algorithm [**?**] is given by

$$\begin{aligned} u &= -k_1\phi_1(\sigma) + z \\ \dot{z} &= -k_2\phi_2(\sigma) \\ \phi_1(\sigma) &= \lceil\sigma\rfloor^{1/2} + \beta\sigma \\ \phi_2(\sigma) &= \frac{1}{2}\lceil\sigma\rfloor^0 + \frac{3}{2}\beta\lceil\sigma\rfloor^{1/2} + \beta^2\sigma \end{aligned} \tag{32}$$

3.3 Finding conditions on gains

Cases

- General case
- State-dependent perturbations and know input gain
- Time-dependent unknown input gain and perturbation
- Closing the loop

3.3.1 General case

Scalar system

$$\dot{\sigma} = \gamma(x,t)u + \varphi(x,t) \tag{33}$$

Assumption 5. *The functions $\gamma(\sigma,t)$ and $\varphi(\sigma,t)$ are Lipschitz continuous functions with respect to t, and $\gamma(\sigma,t)$, $\varphi(\sigma,t) \in \mathcal{C}^1$ with respect to σ.*

Assumption 6. *The uncertain input gain function is bounded by*

$$0 < k_m \leq \gamma(x,t) \leq k_M \tag{34}$$

Assumption 7. *The perturbation term $\varphi(\sigma,t)$ can be split into two components*

$$\varphi(\sigma,t) = \varphi_1(\sigma,t) + \varphi_2(\sigma,t), \tag{35}$$

where is vanishing, i.e. $\varphi_1(0,t) = 0,\ \forall t \leq 0$, and is bounded by

$$|\varphi_1(\sigma,t)| \leq \alpha|\phi_1(\alpha)| \tag{36}$$

Assumption 8. *The total time derivative of $\frac{\varphi_2(\sigma,t)}{\gamma(\sigma,t)}$ is*

$$\frac{d}{dt}\left(\frac{\varphi_2(\sigma,t)}{\gamma(\sigma,t)}\right) = \underbrace{\frac{1}{\gamma}\frac{\partial\varphi_2}{\partial t} - \frac{\varphi_2}{\gamma^2}\frac{\partial\gamma}{\partial t}}_{\delta_1(\sigma,t)} + \underbrace{\left(\frac{1}{\gamma}\frac{\partial\varphi_2}{\partial\sigma} - \frac{\varphi_2}{\gamma^2}\frac{\partial\gamma}{\partial\sigma}\right)}_{\delta_2(\sigma,t)}\dot{\sigma} \tag{37}$$

where $\delta_1(\sigma,t)$ and $\delta_2(\sigma,t)$ are bounded by positive constants

$$|\delta_1(\sigma,t)| \leq \bar{\delta}_1,\ |\delta_2(\sigma,t)| \leq \bar{\delta}_2 \tag{38}$$

Closed-loop system

$$\dot{\sigma} = -k_1\gamma(\sigma,t)\phi_1(\sigma) + \varphi_1(\sigma,t) + \gamma(\sigma,t)\left(z + \frac{\varphi_2(\sigma,t)}{\gamma(\sigma,t)}\right) \tag{39}$$

the system can be broken down

$$\begin{aligned} \dot{\sigma}_1 &= \gamma(\sigma,t)\left(-k_1\phi_1(\sigma_1) + \frac{\varphi_1(\sigma_1,t)}{\gamma(\sigma,t)} + \sigma_2\right) \\ \dot{\sigma}_2 &= -k_2\phi_2(\sigma_1) + \frac{d}{dt}\left(\frac{\varphi_2(\sigma,t)}{\gamma(\sigma,t)}\right) \end{aligned} \tag{40}$$

where $\sigma_1 = \sigma$ and $\sigma_2 = z + \frac{\varphi_2(\sigma,t)}{\gamma(\sigma,t)}$

Theorem 2 (Castillo et al. [**?**]). *Suppose that $\gamma(\sigma,t)$ and $\varphi(x,t)$ of system (33) satisfy (34) and (38). Then, the states σ_1 and σ_2 converge to zero and z converges to $-\varphi(x,t)$, globally and in finite time, if GSTA gains k_1, k_2 and $\beta > 0$ are design as follows for any $\varepsilon > 0$*

$$k_1 > \frac{(1+\varepsilon)}{4\varepsilon k_m}\left(\frac{\alpha}{k_m}(k_M - k_m) + \frac{\bar{\delta}_2\varepsilon}{\beta}(k_M + k_m + 2)\right) + \frac{(1+\varepsilon)}{4\varepsilon k_m}\sqrt{\Lambda_{k_1}} + \frac{\alpha}{k_m} \tag{41}$$

$$\Lambda_{k_1} = \left(\frac{\alpha}{k_m}(k_M - k_m) + \frac{\bar{\delta}_2\varepsilon}{\beta}(k_M + k_m + 2)\right)^2 + 8\varepsilon(k_M - km)\left(\frac{\bar{\delta}_2}{\beta} + \bar{\delta}_1\right) \tag{42}$$

$$k_M\left(2\sqrt{\underline{h}}\epsilon\bar{c} - \frac{\bar{c}}{2}\sqrt{\Delta_{k_2}}\right)^2 + 2\bar{\delta}_1 < k_2 < k_M\left(2\sqrt{\underline{h}}\epsilon\bar{c} + \frac{\bar{c}}{2}\sqrt{\Delta_{k_2}}\right)^2 + 2\bar{\delta}_1 \tag{43}$$

where

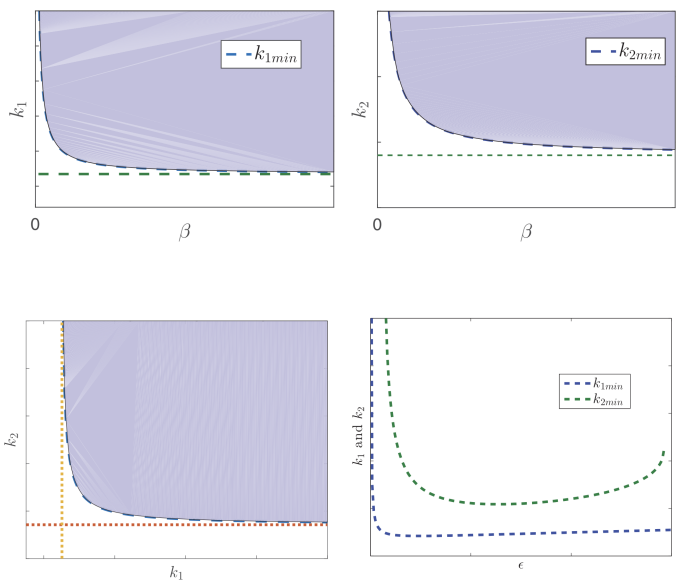
$$\Delta_{k_1} = 16\underline{h}\epsilon - 8\left(\frac{(1+\epsilon)(k_M - k_m)}{\underline{k}_1 k_m}\right) \times \left(\frac{(1+\epsilon)}{\underline{k}_1 k_m}\left[\frac{\bar{\delta}_2\alpha}{\beta} + 2\bar{\delta}_1\right] + \frac{\bar{\delta}_2\epsilon}{\beta} + \frac{\alpha}{k_M}\right),$$

$$\underline{h} = 1 - \frac{\bar{\delta}_2}{\beta}\frac{k_m + \epsilon}{(k_1 k_m - \alpha)},\ \bar{c} = \frac{\underline{k}_1 k_m}{(1+\epsilon)(k_M - k_m)},\ \underline{k}_1 = k_1 - \frac{\alpha}{k_m}$$

Moreover, a system trajectory starting at $\sigma(0) = (\sigma_1(0), |\sigma_2(0))$ reaches the origin in a time smaller than

$$T(\sigma(0)) = \frac{2}{\mu_2}\ln\left(\frac{\mu_2}{\mu_1}V^{1/2}(x_1(0), x_2(0)) + 1\right) \tag{44}$$

Choosing gains



3.4 State-dependent perturbations and know input gain

System

$$\dot{\sigma} = u + \varphi(x,t) \tag{45}$$

Assumption 9. *The function $\varphi(\sigma,t)$ is Lipschitz continuous functions with respect to t, and $\varphi(\sigma,t) \in \mathcal{C}^1$ with respect to σ.*

Assumption 10. *The total time derivative of $\frac{\varphi_2(\sigma,t)}{\gamma(\sigma,t)}$ is*

$$\frac{d}{dt}\left(\varphi_2(\sigma,t)\right) = \underbrace{\frac{\partial \varphi_2}{\partial t}}_{\delta_1(\sigma,t)} + \underbrace{\left(\frac{\partial \varphi_2}{\partial \sigma}\right)}_{\delta_2(\sigma,t)} \dot{\sigma} \tag{46}$$

Closed-loop system

$$\dot{\sigma} = -k_1\phi_1(\sigma) + \varphi_1(\sigma,t) + \left(z + \frac{\varphi_2(\sigma,t)}{\gamma(\sigma,t)}\right) \tag{47}$$

the system can be broken down

$$\begin{aligned} \dot{\sigma}_1 &= \left(-k_1\phi_1(\sigma_1) + \varphi_1(\sigma_1,t) + \sigma_2\right) \\ \dot{\sigma}_2 &= -k_2\phi_2(\sigma_1) + \frac{d}{dt}\left(\varphi_2(\sigma,t)\right) \end{aligned} \tag{48}$$

where $\sigma_1 = \sigma$ and $\sigma_2 = z + \varphi_2(\sigma,t)$

Corollary 1 (1, Castillo et al. [**?**])**.** *Suppose that $\varphi(\sigma,t)$ of system (45) $\delta_1(\sigma,t)$ and $\delta_2(\sigma,t)$ are bounded by $\bar{\delta}_1$ and $\bar{\delta}_2$, respectively. Then, the states σ_1 and σ_2 converge to zero and z converges to $-\varphi(\sigma,t)$, globally in finite-time, if GSTA gains k_1, k_2 and $\beta > 0$ are designed as follows for any $\epsilon > 0$*

$$k_1 > \frac{2(1+\epsilon)\bar{\delta}_2}{\beta} + \alpha \tag{49}$$

$$k_2 > \frac{1}{4\underline{h}\epsilon}\left(\frac{(1+\epsilon)}{(k_1-\alpha)}\left(\frac{\bar{\delta}_2\alpha}{\beta} + 2\bar{\delta}_1\right) + \frac{\bar{\delta}_2\epsilon}{\beta} + \alpha\right)^2 \tag{50}$$

3.5 Time-dependent unknown input gain and perturbation

Scalar system

$$\dot{\sigma} = \gamma(t)u + \varphi(t) \tag{51}$$

Assumption 11. *The functions $\gamma(t)$ and $\varphi(t)$ are Lipschitz continuous functions with respect to t.*

Assumption 12. *The uncertain input gain function is bounded by*

$$0 < k_m \leq \gamma(t) \leq k_M \tag{52}$$

Assumption 13. *The total time derivative of $\frac{\varphi_2(t)}{\gamma(t)}$ is*

$$\delta_1(t) := \frac{d}{dt}\left(\frac{\varphi_2(t)}{\gamma(t)}\right) = \frac{1}{\gamma}\frac{\partial \varphi_2}{\partial t} - \frac{\varphi_2}{\gamma^2}\frac{\partial \gamma}{\partial t} \tag{53}$$

where $\delta_1(t)$ is bounded by positive a constant

$$|\delta_1(t)| \leq \bar{\delta}_1 \tag{54}$$

3.6 Time-dependent unknown input gain and perturbation

Closed-loop system

$$\dot{\sigma} = -k_1\gamma(t)\phi_1(\sigma) + \varphi_1(t) + \gamma(t)\left(z + \frac{\varphi_2(t)}{\gamma(t)}\right) \tag{55}$$

the system can be broken down

$$\begin{aligned} \dot{\sigma}_1 &= \gamma(t)\left(-k_1\phi_1(\sigma_1) + \frac{\varphi_1(t)}{\gamma(t)} + \sigma_2\right) \\ \dot{\sigma}_2 &= -k_2\phi_2(\sigma_1) + \frac{d}{dt}\left(\frac{\varphi_2(t)}{\gamma(t)}\right) \end{aligned} \tag{56}$$

where $\sigma_1 = \sigma$ and $\sigma_2 = z + \frac{\varphi_2(t)}{\gamma(t)}$

Corollary 2 (2, Castillo et al. [**?**]). *Suppose that $\gamma(t)$ and $\varphi(t)$ of system* (51) *satisfy* (52) *and* (54). *Then, the states σ_1 and σ_2 converge to zero and z converges to $-\frac{\varphi(t)}{\gamma(t)}$, globally and in finite-time, if GSTA gains k_1, k_2 and $\beta \geq 0$ are designed as follows for all $\epsilon > 0$*

$$k_1 > \frac{(1+\varepsilon)}{4\varepsilon k_m}\left(\frac{\alpha}{k_m}(k_M - k_m)\right) + \frac{(1+\varepsilon)}{4\varepsilon k_m}\sqrt{\Lambda_{k_1}} + \frac{\alpha}{k_m} \tag{57}$$

$$\Lambda_{k_1} = \left(\frac{\alpha}{k_m}(k_M - k_m)\right)^2 + 16\varepsilon(k_M - km)\bar{\delta}_1 \tag{58}$$

$$k_M\left(2\sqrt{\epsilon}\bar{c} - \frac{\bar{c}}{2}\sqrt{\Delta_{k_2}}\right)^2 + 2\bar{\delta}_1 < k_2 < k_M\left(2\sqrt{\epsilon}\bar{c} + \frac{\bar{c}}{2}\sqrt{\Delta_{k_2}}\right)^2 + 2\bar{\delta}_1 \tag{59}$$

where

$$\Delta_{k_1} = 16\epsilon - 8\left(\frac{(1+\epsilon)(k_M - k_m)}{\underline{k}_1 k_m}\right) \times \left(\frac{(1+\epsilon)}{\underline{k}_1 k_m}2\bar{\delta}_1 + \frac{\alpha}{k_M}\right),$$

$$\bar{c} = \frac{\underline{k}_1 k_m}{(1+\epsilon)(k_M - k_m)}, \ \underline{k}_1 = k_1 - \frac{\alpha}{k_m}$$

3.7 Closing the loop

Example revisited

$$\left(1 + \cos^2(q)\right)\ddot{q} + g\sin(q) + b\left(\dot{q} + \arctan(\dot{q})\right) = u$$

3.8 Key takeaways

Features

- Deals well with matched uncertainties
- Works for uncertain input gain
- Rigorously proves control gain conditions for desired stability properties

Future work

- Combine with adaptation law

4 Adaptive generalized super-twisting algorithm

4.1 Control law

Like in [**?**] the GSTA algorithm is given by

$$\begin{aligned} u &= -k_1\phi_1(\sigma) + z \\ \dot{z} &= -k_2\phi_2(\sigma) \\ \phi_1(\sigma) &= \lceil\sigma\rfloor^{1/2} + \beta\sigma \\ \phi_2(\sigma) &= \frac{1}{2}\lceil\sigma\rfloor^0 + \frac{3}{2}\beta\lceil\sigma\rfloor^{1/2} + \beta^2\sigma \end{aligned} \tag{60}$$

4.1.1 Adaptive gain law

Using same adaptive law as [?]

$$\begin{aligned} \dot{k}_1 &= \begin{cases} \omega_1\sqrt{\frac{\gamma_1}{2}}, & \text{if } \sigma \neq 0, \\ 0, & \text{if } \sigma = 0 \end{cases} \\ k_2 &= 2\epsilon k_1 + \lambda + 4\epsilon^2 \end{aligned} \tag{61}$$

4.1.2 Closed-loop system

$$\dot{\sigma} = -k_1\gamma(\sigma,t)\phi_1(\sigma) + \varphi_1(\sigma,t) + \gamma(\sigma,t)\left(z + \frac{\varphi_2(\sigma,t)}{\gamma(\sigma,t)}\right) \tag{62}$$

the system can be broken down

$$\begin{aligned} \dot{\sigma}_1 &= \gamma(\sigma,t)\left(-k_1\phi_1(\sigma_1) + \frac{\varphi_1(\sigma_1,t)}{\gamma(\sigma,t)} + \sigma_2\right) \\ \dot{\sigma}_2 &= -k_2\phi_2(\sigma_1) + \frac{d}{dt}\left(\frac{\varphi_2(\sigma,t)}{\gamma(\sigma,t)}\right) \end{aligned} \tag{63}$$

where $\sigma_1 = \sigma$ and $\sigma_2 = z + \frac{\varphi_2(\sigma,t)}{\gamma(\sigma,t)}$

Theorem 3 (Borlaug et al. [?])**.** *Suppose that $\gamma(\sigma_1,t)$ and $\varphi(\sigma_1,t)$ in system (33) satisfy Assumptions 1-4, from [?]. Then, the closed-loop dynamics (63) are globally finite-time stable, such that the states σ_1 and σ_2 converge to zero and z converges to $-\varphi(0,t)/\gamma(0,t)$, globally and in finite time, ig the gains k_1 and k_2 are designed as expressed in (61), $\beta > 0, \lambda > 0, \omega_1 > 0, \gamma_1 > 0$, and $\epsilon = \frac{\omega_2}{2\omega_1}\sqrt{\frac{\gamma_2}{\gamma_1}}$, where $\omega_2 > 0$ and $\gamma_2 > 0$.*

Remark 3.
- *The theorem and proof can be extended to n dimensions by separating the problem into n one-dimensional cases*
- *The proof is similar to that of Shtessel et al. [?]*

4.2 Real system

Equations of motion

$$\begin{aligned} \dot{\xi} = J(\eta_2)\zeta &= \begin{bmatrix} R_{Ib}^\top(\eta_2) & 0_{3\times3} & 0_{3\times(n-1)} \\ 0_{3\times3} & J_{k,o}^{-1}(\eta_2) & 0_{3\times(n-1)} \\ 0_{(n-1)\times3} & 0_{(n-1)\times3} & I_{(n-1)\times(n-1)} \end{bmatrix}\zeta \\ M(q)\dot{\zeta} + C(q,\zeta)\zeta + D(q,\zeta)\zeta + g(q,R_{Ib}) &= \tau(q) \end{aligned} \tag{64}$$

where $\xi = \left[\eta_1^\top, \eta_2^\top, q^\top\right]^\top$, here $\eta_1 = [x,y,z]^\top \in \mathbb{R}^{6+(n-1)}$ is the position, $\eta_2 = [\phi,\theta,\psi]^\top \in \mathbb{R}^3$ are Euler angles and $q \in \mathbb{R}^{n-1}$ are the joint angles. The velocity vector $\zeta = \left[\nu^\top, \omega^\top, \dot{q}^\top\right]^\top \in \mathbb{R}^{6+(n-1)}$ comprises body-fixed linear and angular vehicle velocities ν and ω, and joint velocities $\dot{q}$.

4.2.1 Tracking control law using AGSTA

We choose sliding manifold

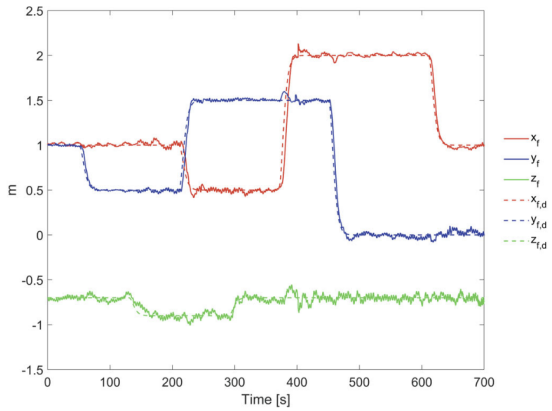
$$\sigma = \zeta - \zeta \in R^{6+(n-1)} \tag{65}$$

and the control input is given by

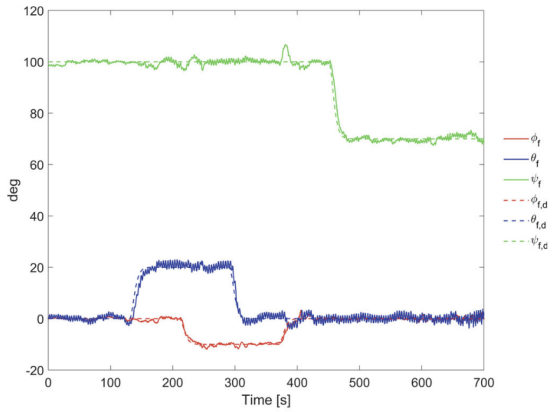
$$\tau(q) = u_{\text{AGSTA}} \in R^{6+(n-1)} \tag{66}$$

4.3 Experiments

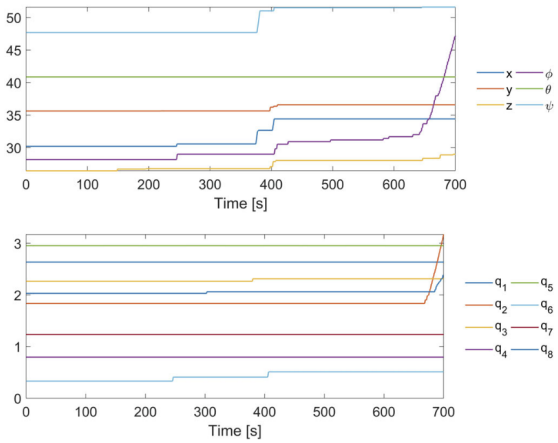
Tracking performance [?] Gain evolution [?]



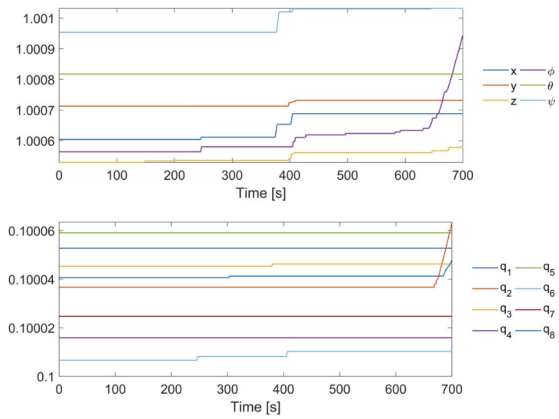
(A) Front end position



(B) Front end orientation



(A) Evolution of $k_1(t)$ over time



(B) Evolution of $k_2(t)$ over time

4.4 Comparison

Performance [?]
Actuation [?]

4.5 Key takeaways

Features

- Deals well with matched uncertainties
- Works for uncertain input gain
- Practically proven control method with desirable stability guarantees
- AGSTA demonstrates better tracking performance with GSTA with constant gains

Future work

- Coupled MIMO formulation
- Adaptive law with fewer parameters
- Adaptive law that enables decreasing controller gain

Algorithm		RMSE		Maximum error	
		Task 1	Task 2	Task 1	Task 2
AGSTA	Position	0.0521	0.0760	0.3323	0.4011
	Orientation	0.0220	0.0296	0.1040	0.1054
STA	Position	0.0651	0.0839	0.4213	0.5638
	Orientation	0.0235	0.0321	0.1253	0.1831
GSTA	Position	0.0649	0.0915	0.4154	0.5304
	Orientation	0.0215	0.0328	0.1267	0.1560
PID	Position	0.0851	0.1020	0.4900	0.5545
	Orientation	0.0263	0.0366	0.1394	0.1797

	RMS			
Algorithm	τ_{thr}	$\dot{\tau}_{thr}$	τ_q	$\dot{\tau}_q$
AGSTA	5.0325	7.8897	2.1488	0.3898
STA	3.6394	5.7950	2.2093	0.3645
GSTA	3.6527	3.3307	2.2126	0.3398
PID	3.2469	42.1036	2.1960	5.3478

5 Multivariable generalized super-twisting algorithm

5.1 Problem statement

5.1.1 Multivariable system

Consider the multivariable system

$$\dot{x} = f(t,x) + G(t,x)u \tag{67}$$

where $x \in \mathbb{R}^n$ is the state vector, $u \in \mathbb{R}^n$ is the contorl input, $f : \mathbb{R}_+ \times \mathbb{R}^n \to \mathbb{R}^n$ is an uncertain vector field containing uncertainties and/or perturbations, and $G : \mathbb{R}_+ \times \mathbb{R}^n \to \mathbb{R}^{n\times n}$ is the uncertain input matrix.

5.1.2 Input matrix

The uncertain input matrix can be rewritten

$$G(t,x) = (I + \Delta_G(t,x))\, G_0(t,x) \tag{68}$$

where $G_0 : \mathbb{R}_+ \times \mathbb{R}^n \to \mathbb{R}^{n\times n}$ corresponds to the nominal part, which is assumed to be invertible, and $\Delta_G : \mathbb{R}_+ \times \mathbb{R}^n \to \mathbb{R}^{n\times n}$ is the uncertain term

5.2 Multivariable super-twisting algorithm

5.2.1 Control law by Moreno et al. [?]

The MGSTA is given by

$$\begin{aligned} u &= -k_1\phi_1(x) + bG_0^{-1}(t,x)v \\ \dot{v} &= -k_2\phi_2(x) \end{aligned} \tag{69}$$

The nonlinear functions $\phi_i : \mathbb{R}^n \to \mathbb{R}^n$ are monotonically increasing, ϕ_1 is continuous everywhere and ϕ_2 is continuous everywhere except possibly at $x = 0$.

Nonlinear functions

$$\phi_1 = \left(\alpha|x|^{-p} + \beta\right) x \tag{70}$$

$$\phi_2 = J(x)\phi_1(x) = c(x)\phi_1(x) \tag{71}$$

5.2.2 Decomposed perturbations

The uncertain term $f(t,x)$ is assumed to be decomposed as

$$f(t,x) = f_1(t,x) + f_2(t,x) = \Delta_1(t,x)\phi_1(x) + f_2(t,x) \tag{72}$$

where the function $f_2(t,x)$ is such that

$$\frac{d}{dt}\left[\left(b\left(I+\Delta_G\right)^{-1} f_2\right)\right] = \Delta_2(t,x)\phi_2 + \Delta_3(t,x)\dot{x} \tag{73}$$

where $\Delta_i : \mathbb{R}_+ \times \mathbb{R}^n \to \mathbb{R}^{n\times n}, i = 1,2,3.$

For the dynamical system (67) and controller (69), (70) and (71), the following conditions hold for all $t \geq 0, x \in \mathbb{R}^n$.

Assumption 14. *There exit constants $g_M > 0, \delta_g \geq 0$ and $g_m^\Delta > 0$ such that*

$$\begin{aligned} |G(t,x)| &\leq g_M \\ |\Delta_G(t,x)| &\leq \delta_g \\ Sym\left\{G(t,x)\right\} &\geq g_m^\Delta I \end{aligned} \tag{74}$$

Assumption 15. *Matrices $\Delta_i, i = 1,2,3,$ in (72) and (73), are bounded, there exist $\delta_i \geq 0$ such that*

$$|\Delta_i(t,x)| \leq \delta_i, i = 1,2,3.$$

Assumption 16. *There exist positive constants $0 < \gamma_1^\Delta \leq \gamma_2^\Delta$ such that*

$$0 < \gamma_1^\Delta I \leq Sym\left\{\mathcal{J}(x)\left(I+\Delta_G\right)\right\} \leq \gamma_2^\Delta I \tag{75}$$

As a consequence of these assumptions, there also exist constants $\gamma_3, \gamma_4, \gamma_5 > 0$, and $0 < \gamma_1^\Delta \leq \gamma_2^\Delta$ such that

$$\begin{aligned} Gram\left\{\mathcal{J}G\right\} &\leq \gamma_3 I \\ 2Sym\left\{\Delta_G \mathcal{J}G\right\} &\leq \gamma_4 I \\ Gram\left\{\Delta_G^\top\right\} &\leq \gamma_5 I \end{aligned} \tag{76}$$

Closed-loop dynamics The closed-loop dynamics including MGSTA is given by

$$\dot{x} = f_1 + \left(I+\Delta_G\right)\left(-G_0 k_1 \phi_1 + bz\right) \tag{77}$$

$$\dot{z} = -k_2\phi_2 + \Delta_2\phi_2 + \Delta_3\dot{x} \tag{78}$$

Theorem 4 (Moreno et al. [**?**]). *Select $\alpha > 0, \beta > 0, b > 0$ and $p \in \left(0, \frac{1}{2}\right]$ and let Assumption 1 be satisfied. Assume further that the inequality*

$$\gamma_1^\Delta g_m^\Delta > \gamma_4 + 2\sqrt{\gamma_3\gamma_5} \tag{79}$$

is satisfied. Then, for arbitrary values of $\delta_2, \delta_3, \delta_3$, there exist positive constant controller gains k_1, k_2 such that $\left[\sigma^\top, z^\top\right]^\top$ is globally robust finite-stable.

5.2.3 Algorithm 1

For proper selection of the control gains k_1 and k_2 consult the following algorithm

1. Select $\alpha > 0, \beta > 0, b > 0$ and $p \in \left(0, \frac{1}{2}\right]$. The parameter p should be selecting on the size of the perturbations.

2. Select $p_2 = p_2^*$ and $p_1 = p_1^*$ such that
$$\gamma_1^\Delta > p_2\mu_4 \tag{80}$$
and
$$\gamma_1^\Delta g_m^\Delta - \gamma_4 - \Gamma_0(p_1,p_2) > 0 \tag{81}$$
are satisfied for $p_2 = p_2^*$ and $p_1 \geq p_1^*$. For proper selection of the control gains k_1 and k_2 consult the following algorithm

3. Find a value $p_1^{\#}$ of p_1 such that $p_1 \geq p_1^{\#}$ and

$$\Gamma_1(p_1, p_2)p_1 > \Gamma_2(p_1, p_2) \tag{82}$$

is satisfied for every $p_1 \geq p_1^{\#}$. This is always feasible. For proper selection of the control gains k_1 and k_2 consult the following algorithm

(a) Fix $p_2 = p_2^*, p_1 \geq p_1^{\#}$ and $p_1 p_2 > 1$, choose

$$k_2 = b\frac{p_1}{p_2}, \tag{83}$$

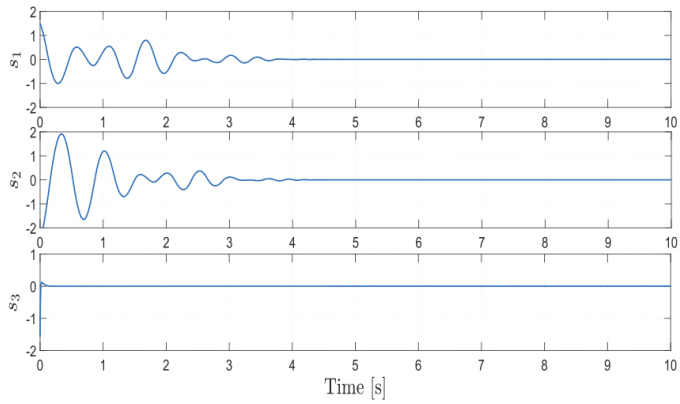
and

$$\frac{-\alpha_1 - \bar{\alpha}}{2\alpha_2} < k_1 < \frac{-\alpha_1 + \bar{\alpha}}{2\alpha_2} \tag{84}$$

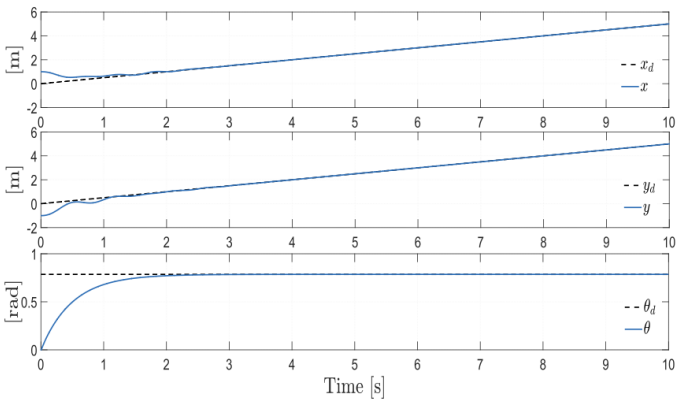
with $\bar{\alpha} = \sqrt{\alpha_1^2 - 4\alpha_2\alpha_0}$

5.3 Example: Four-wheeled omidirectional robot

Sliding variables, [**?**]



Position and orientation, [**?**]

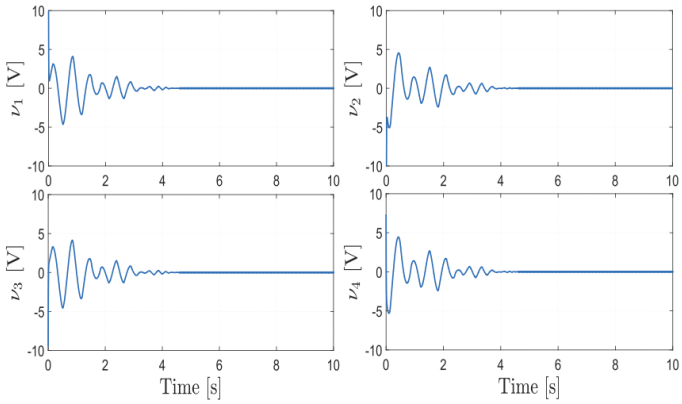


Armature voltages, [**?**]

5.4 Key takeaways

Features

- Deals well with matched uncertainties



- Works for uncertain input gain
- Coupled MIMO formulation

Future work

- Implement adaptive law
- Cleaner formulations and communication